



Notes

Jul 29, 2026

Meeting Jul 29, 2026 at 12:27 UTC

Meeting records  Transcript

Summary

Meeting reviewed platform technical architecture and bioinformatics modeling strategies for prioritizing computational bacterial infection candidates.

Bioinformatics Methodology Standards

Standardized scientific nomenclature for bacterial strains ensures accuracy across structural models. AlphaFold serves as the preferred structural modeling tool for globular proteins.

Platform Capabilities Discussion

The interface visualizes extensive ligand docking data and enables researchers to identify promising candidates for validation. A holistic bird's-eye approach defines the unique platform value proposition.

Modeling Refinements and Normalization

Consensus scoring and data normalization address inconsistencies in priority rankings across different computational tools. The team decided to exclude ESM fold integration to maintain model focus.

Decisions

Aligned

- **PubChem compound hyperlinks chosen** Compound names in the UI are set to include hyperlinks redirecting users to the PubChem database for detailed information.
- **Metric tooltip popups approved** UI design will incorporate hover-over tooltips that display a descriptive paragraph regarding the scientific significance of metrics such as Vina affinity, bond energy, and confidence scores.
- **NCBI sequence redirection finalized** Gene sequence references are set to link directly to the NCBI website, replacing the manual inclusion of strain numbers for sequence retrieval.
- **AlphaFold tool prioritization confirmed** AlphaFold is confirmed as the primary structural prediction tool, with the team deciding that implementation of ESM fold is not necessary at this stage.
- **Interaction view customization strategy set** UI implementation will feature a checkbox system allowing users to toggle between a default non-covalent interaction view and a customized view that includes covalent interactions.

We've **updated the Decisions section** using your feedback.

Let us know what you think: [Helpful](#) or [Not Helpful](#)

Next steps

- ☐ [Ayush Singh] Send Link: Provide the meeting link in the designated group.
- ☐ [Ayush Singh] Format Nomenclature: Update bacterial names to adhere to scientific standards, including italics, correct capitalization, and inclusion of accession numbers.
- ☐ [Ayush Singh] Add Compound Hyperlinks: Create links on compound names that redirect users to their corresponding pages on PubChem.

- ☐ [Ayush Singh] Implement Tooltips: Develop hover based popups that provide descriptive explanations for metrics like Vina affinity and hydrogen bond significance.
- ☐ [Ayush Singh] Add Sequence Hyperlinks: Incorporate links on gene sequences that lead directly to the NCBI website for further details.
- ☐ [Ayush Singh] Clarify Receptor Labels: Investigate and explain the meaning of receptor chain labels currently displayed in the model outputs.
- ☐ [The group] Validate Data: Conduct a thorough review and validation of all generated docking data before finalizing new scoring algorithms.
- ☐ [Avishek Banik] Create Word Document: Draft the materials and methods document based on project details to ensure team alignment.
- ☐ [Ayush Singh] Share Email: Provide email address in the WhatsApp group for document sharing.
- ☐ [Ayush Singh] Create Meeting Summary: Summarize the meeting notes into 5 to 10 points immediately.
- ☐ [Avishek Banik, ANIRBAN SHAOO] Evaluate Interaction Models: Assess the necessity of including covalent and electrostatic interactions as a checkbox option for the tool.
- ☐ [ANIRBAN SHAOO] Verify Phytochemical Interactions: Check literature to confirm if phytochemicals utilize covalent interactions.
- ☐ [Avishek Banik] Provide Implementation Feedback: Confirm if the checkbox option for covalent interactions should be implemented.

Details

- **Meeting Logistics and Coordination:** Participants began the meeting by addressing logistical difficulties regarding the calendar invitation and ensuring everyone was included in the session across different time zones.
- **Bacterial Nomenclature and Standardization:** Avishek Banik emphasized the necessity of using correct scientific naming conventions for bacteria, specifically noting that names must be in italics, with the genus capitalized and

the species in lowercase, followed by the specific strain or National Center for Biotechnology Information accession number ([00:00:57](#)).

- **Strain Significance in Microbiology:** The group discussed the importance of strain identification, noting that while bacteria may be genetically similar, their metabolic responses differ based on environmental factors like temperature and humidity ([00:03:08](#)).
- **Data Visualization and Metrics:** Ayush Singh presented the current state of the user interface, which includes analysis of 12 targets across three different bacteria and 24 ligands. The team reviewed scoring metrics, including the vina affinity score and the dev doc score ([00:04:45](#)) ([00:06:33](#)).
- **Role of AlphaFold in Structural Analysis:** The discussion highlighted the industry-wide shift toward using AlphaFold for creating structural models of globular proteins, confirming that it is currently the preferred tool for the project ([00:05:35](#)).
- **User Interface Enhancements for Scoring:** Avishek Banik requested that the interface be improved with hyperlinks to the PubChem database for compound information and hover-over popups to explain the significance of metrics like vina affinity, helping laypeople interpret the data ([00:07:24](#)).
- **Interpreting Computational Results:** Nishant Agrawal raised concerns about how to present the data to users, specifically regarding the interpretation of priority scores and the goal of demonstrating which compounds are the top candidates for microbial infection ([00:10:00](#)).
- **Biofilm Challenges and Wet Lab Validation:** Avishek Banik explained that while genetic targets are easier to study in silico, biofilms are difficult to model computationally because their formation is environmentally dependent. The team agreed that the current results serve as candidates for future wet lab validation, which would then be used to train and refine the data model ([00:11:59](#)).
- **Strategic Grant Proposals:** The team discussed how to frame their work for grant reviewers, emphasizing that their computational approach narrows down a large number of potential products to a few highly probable candidates, thereby streamlining the process for researchers doing wet lab experiments ([00:15:12](#)).

- **Data Management and File Structure:** Ayush Singh provided an overview of the data architecture, noting that they have cached results for every docking pair, totaling over 288 combinations and more than 2,000 files, including various post-docking outputs ([00:17:02](#)).
- **Refining UI Details:** The team reviewed the interaction parser and the visual knowledge graph within the interface. Discussions arose regarding the labels for receptor chains and the need for Ayush Singh to clarify what specific residue positions and chain identifiers represent ([00:18:55](#)).
- **Model Refinement and Data Science:** Nishant Agrawal indicated that they would perform deeper data science work on the encoder-decoder models to improve the priority scoring system, with input from the rest of the team once the initial data is validated ([00:21:45](#)).
- **Publication Strategy:** Avishek Banik suggested that the team should aim to publish their findings, modeling their process after the AlphaFold publication, which involved years of similar work before release ([00:22:34](#)).
- **Consistency and Hallucination Concerns:** Anirban Sahoo expressed concern about the inconsistency of results across different targets, where top priority compounds frequently changed, leading to a discussion on whether the model was producing unreliable output ([00:23:55](#)).
- **Consensus and Normalization Strategies:** The group explored the possibility of creating a "consensus" score by averaging results from different bioinformatics tools. Nishant Agrawal and Avishek Banik discussed the need to normalize these different units of measurement to arrive at a unified priority ranking ([00:26:28](#)).
- **Future Normalization Methodology:** The team discussed the conceptual difference between internal standard normalization for wet lab experiments and the current reliance on machine calibration, planning to include this methodological nuance in their future documentation ([00:31:07](#)).
- **Overall Platform Feedback:** Avishek Banik provided positive feedback on the work performed by Ayush Singh and Anirban Sahoo, noting that the platform is a significant achievement, though it requires wet lab validation to reach completion ([00:34:50](#)).

- **Establishing Platform Uniqueness:** The team discussed how to articulate the "novelty" of their platform. They concluded that while the individual bioinformatics tools are not new, the holistic "bird's-eye view" approach of combining multiple targets and ligands to draw specific, actionable inferences creates a unique research value proposition ([00:36:59](#)).
- **Documentation and Knowledge Transfer:** Avishek Banik directed the team to prepare a document outlining the materials and methods used, which will be shared among the group to ensure everyone is aligned for the next phase of work ([00:39:34](#)).
- **Expanding Platform Scope:** The group discussed the potential to extend the protein-ligand interaction platform beyond the current bacteria to other areas, such as cancer research or Fusarium, as the underlying technology is versatile ([00:42:13](#)).
- **Tool Selection for Future Steps:** The team agreed that their current use of AlphaFold is sufficient, and they decided not to pursue integrating ESM fold at this time, as it would not necessarily improve upon their current structural predictions ([00:44:07](#)).
- **Interaction Modeling Considerations:** Anirban Sahoo suggested investigating whether covalent interactions should be included alongside non-covalent interactions. The group decided to explore adding a user-selectable checkbox for these interactions, allowing users to toggle between default and customized modes ([00:44:58](#)).
- **Closing Logistics:** The team confirmed their intention to continue refining the project, with Nishant Agrawal planning a trip to Kolkata to meet the team and finalize the presentation deck for future pitches ([00:49:10](#)).

You should review Gemini's notes to make sure they're accurate. [Get tips and learn how Gemini takes notes](#)

How is the quality of **these specific notes**? [Take a short survey](#) to let us know your feedback, including how helpful the notes were for your needs.



Transcript

Jul 29, 2026

Meeting Jul 29, 2026 at 12:27 UTC - Transcript

00:00:00

Ayush Singh: email that you have joined with.

Nishant Agrawal: No no no no because we are not part of the invite you send the instant meeting so we will not get the email you need to send it

Ayush Singh: Oh, okay. Okay. I'll send it in the group.

Nishant Agrawal: separate I'm doing I'm doing this every

Ayush Singh: No.

Nishant Agrawal: day 10 times at least with all time zones of

Avishek Banik: Yeah. Yeah. Since since morning 9:00 I'm sitting in front of uh like uh laptop just

Nishant Agrawal: Oh my

Avishek Banik: 5:30 it got finished and again I have started at 6:00.

Nishant Agrawal: god.

Avishek Banik: So for me also and from 7 like 6:45 onwards I have another meeting.

Nishant Agrawal: Oh my god.

Avishek Banik: No, it's it's a like I'm it's feeling irritating the whole day in front of like laptop I'm sitting over here.

Nishant Agrawal: But it's good actually mentally.

Avishek Banik: It's I'm not enjoying it's it's it's sometime it feels too

Nishant Agrawal: Anyway,

Avishek Banik: much.

Ayush Singh: Yes, sir. I'll show the

00:00:57

Nishant Agrawal: shalush.

Ayush Singh: screen. Is my screen visible?

Avishek Banik: It's on the same UI now.

Ayush Singh: Yeah, I updated the UI. Is everything? So, we had we had 12 targets.

Avishek Banik: Okay.

Ayush Singh: Oh, so there were four targets each. So these are the three bacterias.

Avishek Banik: Yeah.

Ayush Singh: These are the three different

Avishek Banik: So first first see it's a it's a bacteria. So there are certain nomenclature things are there.

Ayush Singh: bacteria.

Avishek Banik: Uh do do so be scientific. *Sudonusa* on one you have to take care of these things. So P will be in capital S onwards all will be small. A will be small and after that P will be in caps except

Ayush Singh: Okay. Let's

Avishek Banik: caps all will be small and this *sudonusa* it will be on italics and

Ayush Singh: ensure

Avishek Banik: if any strain are there you need to mention the strain number also or in bracket mention its accession number NCBI accession number.

Ayush Singh: strain as in

Avishek Banik: Strain means look Aushing he's from Mumbai another Singh clone is

00:02:05

Ayush Singh: Yeah.

Avishek Banik: there maybe it's in West Bengal so it's Aushing strain Mumbai and

Ayush Singh: Okay.

Avishek Banik: Aayushing strain Kolkata it's like that so there are different kinds of sudomonus are there and their origin is different so that is called the strain so all the

Ayush Singh: Okay.

Avishek Banik: bacterial name will be like that safy oras will be capital rest will be all small and it all will be italics and mention the strain number or the accession number. If you don't know the strain, if you have any confusion, ask us or put the uh accession number. Accession number is given to a particular thing, particular sequence. So, it's much more specific. Okay.

Nishant Agrawal: Yeah,

Ayush Singh: Okay.

Avishek Banik: Okay.

Nishant Agrawal: these comments are also logging into the Gemini also I use. So just remember the pointers and work with an to close this You

Ayush Singh: Uhhuh.

Avishek Banik: things things like that look sudomona one is isolated from Kolkata another

Nishant Agrawal: got

Avishek Banik: is isolated from maybe Kashmir so the thing is that they are genetically genetically

00:03:08

Ayush Singh: Mhm.

Avishek Banik: same but think about the temperature humidity all the thing in Kolkata and it's in

Ayush Singh: Mhm. Yeah. Exactly.

Avishek Banik: in Kashmir so they are metabolically different that that is why the concept of

Ayush Singh: Yeah.

Avishek Banik: strain came in Okay,

Ayush Singh: Okay.

Avishek Banik: they they genetically same but they are metabolically different.

Nishant Agrawal: heat.

Ayush Singh: Okay.

Avishek Banik: So that's why different strains in microbiology are there

Ayush Singh: But but sir when we tested for the docking like this docking that we did on the AOS compound

Avishek Banik: yeah

Ayush Singh: so that would be like what did we try it on like this? I just tried it on this target.

Avishek Banik: strain specific. No the no no no no no no no

Ayush Singh: So we currently here we don't have any specific things. No currently we just use the targets.

Avishek Banik: no. It's not like that. So it's a target target of pseudomona ceruogenosa.

Ayush Singh: Mhm. Mhm. Mhm.

Avishek Banik: So you took the sequence from somewhere.

00:03:56

Avishek Banik: So mention the accession number. So if you take that history you'll you'll reach up to the strain.

Ayush Singh: Mhm.

Avishek Banik: So here you need to mention the mention the accession number like the aminosid sequence from where you are getting the amino acid sequence. There

must be some origin. There must be some accession number. So mention that accession number over there.

Ayush Singh: Okay.

Avishek Banik: Okay.

Ayush Singh: Okay.

Avishek Banik: It's a strain specific one.

Ayush Singh: Okay.

Avishek Banik: You need to look for these things. These are small things but these

ANIRBAN SHAOO: Um uh yeah um yes sir I I have already given him the data

Avishek Banik: matter.

ANIRBAN SHAOO: set of that I'll tell him what to add

Nishant Agrawal: What's

Avishek Banik: Yeah, it's it's all right.

Nishant Agrawal: this?

Avishek Banik: But whenever you are looking up looking to UI see for example any scientist or someone if we are going

ANIRBAN SHAOO: actually

Avishek Banik: to present these are not scientific term and when look he engineer please do not expect to like learn these basic things.

00:04:45

ANIRBAN SHAOO: yeah I will tell him I will tell him I'll tell him Okay.

Avishek Banik: These are ABCs. Okay. Okay. Okay. Okay. I was Go

Ayush Singh: Yeah sir. So yeah these are the three different bacteria that we worked with and the four targets

Avishek Banik: ahead.

Ayush Singh: each for all the three bacteria. So we tested it among all the 24 lians that we had. For example, for PQSR, if you see PQSR, the highest score came in with

imperatonian torine and the score that we got was 54.5. It's our evidence score like the priority score that we got. And for

Avishek Banik: Yes. I want to I just just wait.

Ayush Singh: the

Nishant Agrawal: That is a that is not a important score.

Avishek Banik: I I want to see

Nishant Agrawal: Vishek just to add it.

Avishek Banik: this.

Ayush Singh: Mhm.

Nishant Agrawal: We will we will change that score that but more than that score you will focus on this hydrogen bonds div do and option just to just to like

00:05:35

Avishek Banik: Yes. Yes. Yes. Yes.

Nishant Agrawal: yeah

Avishek Banik: for like protein uh like structuring and all this. The whole world is currently moving to alpha fold.

Nishant Agrawal: see alpha fold is for the uh mostly like very good on the target like what Aush also used right Aush for that instruction. structure creation,

Ayush Singh: Yes.

Nishant Agrawal: right?

Avishek Banik: structural it's a it's a structural yeah if you have a globular protein if you have

Nishant Agrawal: He's doing that, right? He's doing that.

Ayush Singh: Stop.

Avishek Banik: a globular protein sequence so alpha fold currently it's it's like it's a nature preper in 2021 and after that like the whole world has jumped to the alpha

Nishant Agrawal: So yeah,

Avishek Banik: fold

Nishant Agrawal: they use physics by the way inside their model. This is not uh not just AI. They use a heavy physics on that.

Avishek Banik: Yes, it should be because it's a it's called Ramachandan plot. So if you have a like sequence,

Nishant Agrawal: Yes,

Avishek Banik: it's it's a basically bioisics and certain chemistry parts are there.

00:06:33

Nishant Agrawal: they use partial differential equation inside

Avishek Banik: Yeah. So many things are there because if two mites are there,

Nishant Agrawal: it

Avishek Banik: how they are interacting, how they are like forming the bond and what is the bond energy?

Nishant Agrawal: gone dynamics.

Avishek Banik: So, so many things uh that Yeah. Yeah. Yeah.

Nishant Agrawal: So we've built the model right by the way I showed you in the Kolkata disease progression is based on bioisics.

Avishek Banik: Yes. Yes. Yes. Yes.

Nishant Agrawal: Okay. Yeah. Go ahead. You explain

Avishek Banik: Yes.

Ayush Singh: Yeah. So here you get all the different scores like the vina affinity score and the dev doc score and

Nishant Agrawal: it.

Ayush Singh: for like uh these four targets we can like when we click on a specific target we get the number one candidate out of all that lians that we had. See here the number one score we got was from oxyolin A. So we got this score from that. Similarly for if we try for PL we get

00:07:24

Avishek Banik: Okay. Okay. I was I was certain things are coming to my mind.

Ayush Singh: nimolide.

Avishek Banik: Uh can you like uh these compounds can you do certain

Ayush Singh: Mhm.

Avishek Banik: hyperlink so that it will go to pup or like that?

Ayush Singh: Hyperlink as in

Avishek Banik: Hyperlink means this nim nimbo light nimolide if you click

Ayush Singh: Yeah.

Avishek Banik: limbide it will go to the popc where nimolites are there

Ayush Singh: Uhhuh. Yeah. Yeah. We can do that. Yeah. It's It's easy.

Avishek Banik: uh this one yeah uh this is one thing and another

Ayush Singh: That's not That's not not an issue. We can add that link to that.

Avishek Banik: thing was coming to my mind uh okay go ahead if if it's coming then I'll again

ANIRBAN SHAOO: Something regarding

Ayush Singh: Should I add it in the targets as well?

Avishek Banik: little

ANIRBAN SHAOO: this.

Ayush Singh: Like I can add it to the nimble the lians. I can add that to the target as well if you

00:08:17

Avishek Banik: and another thing go down go down go down go down go down go

Ayush Singh: want.

Avishek Banik: below. Yeah yeah yeah yeah. So so here here here here here come to compound name. So if you if you do this this thing it will it will automatically be diverted to pap and here vina affinity kilo calalorie per mole kilo calalorie per

Ayush Singh: Okay. Mhm.

Avishek Banik: mole shahu on it's not written in in scientific format so check this this thing up and then what does it signify so so if certain people are like putting your c shirt over vina affinity it will show some information can you do like that. So when you remove the card shot it will it will go it will go up it will not be displayed.

Nishant Agrawal: Okay.

Avishek Banik: So what is the significant of the vina affinity? Write a paragraph. What is the significance of this different uh div do confidence? What is the significance of age bond like that? Can you do that?

00:09:18

Avishek Banik: Small informations. Small informations. If people are dragging the cutart over here,

Nishant Agrawal: Yeah.

Avishek Banik: it will be displayed.

Ayush Singh: Okay.

Avishek Banik: And if you remove the cutart,

Ayush Singh: Okay.

Avishek Banik: uh that display will go. Can we do that?

Ayush Singh: Okay. Yeah.

Nishant Agrawal: So when you mouse your mouse,

Ayush Singh: Like Yeah. When the cursor will be on that when infinity it will show a small paragraph.

Nishant Agrawal: there's a popup. Popup is up.

Avishek Banik: Yes.

Ayush Singh: Yeah.

Avishek Banik: Yes. Yes. Significance.

Nishant Agrawal: So yes

Avishek Banik: So a layman can also understand what you have

Ayush Singh: That's right. That is

Nishant Agrawal: another also which I was asking uh Aayush also to day back that

Avishek Banik: done.

Ayush Singh: right.

Nishant Agrawal: uh what is the how do we give some message to the user who's seeing this see there are some numbers are coming autod talk venas all are different numbers our if you remove our priority score how

Avishek Banik: Yeah.

00:10:00

Nishant Agrawal: do we present this number which is bad good and all the correlation how do we do that That's what is not coming out for me. What do you think?

Avishek Banik: Um yes. Uh

Nishant Agrawal: Like I wish people will ask right okay as we also ran this but uh what is the

Avishek Banik: so

Nishant Agrawal: interpretation we are giving to anybody not just anybody.

Avishek Banik: means uh you want uh like one interpretation out of all this

Nishant Agrawal: Yeah, means why this activity we are doing and let's say we are doing uh like uh we have the

Avishek Banik: Okay.

Nishant Agrawal: like every components and we have that virus components right and uh we are trying to do some docking and other stuff right but uh what is the message that we are trying to deliver out of

Avishek Banik: Yeah. Here we are trying to deliver that what are the probable drugs that

Nishant Agrawal: this

Avishek Banik: I compounds can be given against that particular microbial infection. So, so, so what we can do here there will be some consens consensus thing like

00:10:59

Nishant Agrawal: Okay.

Avishek Banik: say uh these top five compounds. So these top five compounds will be in our display that these are the top hits that can be used again. These this these microorganisms

Nishant Agrawal: So on what basis we will decide that only between these three components or we need some more analysis. That is my Okay.

Avishek Banik: we can we can add we can add definitely the more you will add it will

Nishant Agrawal: Okay. So let's finish this. But yeah,

Avishek Banik: be much more like Yeah.

Nishant Agrawal: even the Swiss guys will ask this uh for this uh even I was talking to an Italian guy two days back and he just asking me this type of questions very

Avishek Banik: Yeah,

Nishant Agrawal: straight.

Avishek Banik: we can add so many look the more the more you can add it will be much more like foolproof thing will will much more closer to the actual thing.

Nishant Agrawal: Dub.

Avishek Banik: You can

Nishant Agrawal: like the some we can show that right it is trying to attack the bofilm that you are seeing right that day when I

00:11:59

Avishek Banik: add

Nishant Agrawal: was in Kolkata that uh how do we show some proof

Avishek Banik: currently. Yeah. Currently we are working on on gene level. If certain genes are uh look uh these this biofilm and gene is completely different. Bofilm basically it's formed by the bacteria.

Nishant Agrawal: Ah,

Avishek Banik: You do not have a control over the formation of the bofilm. At genetic level what you can you can work with the protein we need the structure

Nishant Agrawal: okay.

Avishek Banik: but the biofilm it's it's not in our hand how it will form the threedimensional structure we cannot do

Nishant Agrawal: No, I was it can be anything.

Avishek Banik: that

Nishant Agrawal: I just took the bioilm because it was struck my mind there. But anything we can show

Avishek Banik: protein protein we can predict like see for say for example a protein amino acid at

Nishant Agrawal: which

Avishek Banik: certain sequence if it folds the folding pattern will be same. Okay.

Nishant Agrawal: Okay. Okay.

Avishek Banik: But but the problem is that the biofilm if it it form 3D structure depending upon the environment the 3D structure will form and we do not have any kind of hand over the bofilm but but the protein sequence these are much more easy to target.

00:13:13

Nishant Agrawal: Uh

Avishek Banik: Boflim here we cannot target. So bioflim how how we can target we have to do wet lab experiment.

Nishant Agrawal: so we can uh so this can also tell that this is a candidate for wetl. Like if you see the last two columns.

Avishek Banik: Definitely definitely and then and then and then when we have the weightlap

Nishant Agrawal: Yeah.

Avishek Banik: data we can train this the data set that with this combination the weightlap thing are also working.

Nishant Agrawal: Yeah.

Avishek Banik: So that is the second part when we we have the fund yeah validation part.

Nishant Agrawal: That is the lending part. Validation part. Yeah.

Avishek Banik: So when we have the funds so we will put we will do the validation and there are several values will be there for biohail disruption and xyz and then we will train our this

Nishant Agrawal: Yes.

Avishek Banik: data set with the actual real world bioh data set and then we have the complete picture. So this is the half part. It can go in either way.

00:14:09

Avishek Banik: But that experiment what we will do in in vitro it is the last thing. Maybe certain things we cannot predict or certain things we cannot say with the with this bionformatics. Certain things will be there that is unexplainable. We cannot explain the result. But you have to obey that actual result. Say for example one compound here it's showing the highest affinity but there you will see that top four out of top four the fourth one is showing highest affinity. So that is the catch. We need to drag down the differences between these one to four. And when we have the actual data when we can train our this thing with actual data things will be complete.

Nishant Agrawal: completely different.

Avishek Banik: completely different for that

Nishant Agrawal: Yeah, I think that should we should do actually. That's a good idea point actually.

Avishek Banik: you need to give us money. Yeah.

Nishant Agrawal: Absolutely.

Avishek Banik: Funds for without funds. So the these are all virtual. So without funds we cannot enter into the

00:15:12

Nishant Agrawal: Yes. Yeah. No,

Avishek Banik: system.

Nishant Agrawal: this one in this process whatever we are interpreting with this right in order to get the fund right to showcase to these

Avishek Banik: Yeah. Yeah.

Nishant Agrawal: guys how we can make an impact with this or maybe we can run some ESM fold or some folding predictions.

Avishek Banik: that that that you can do that you can you can like pull everything down.

Nishant Agrawal: Huh?

Avishek Banik: But if I am the uh reviewer of this this kind of grant, I will ask where where is the weightlap data? Have you done the

Nishant Agrawal: No,

Avishek Banik: experiments?

Nishant Agrawal: so that anyway we are saying right that for that we are pulling and proposal making a proposal that we need to get

Avishek Banik: Yeah.

Nishant Agrawal: that but in order to convince that that this is what is the path that we wanted to go and it is based

Avishek Banik: Yes. Yes. So what what it will do it will narrow down lots of from lots of product to a

00:16:00

Nishant Agrawal: on your

Avishek Banik: few product so that yeah whoever is doing the wet lab experiment it will make his or her life easy and it will give the doctor at a particular direction that yes this is the way to go ahead.

Nishant Agrawal: Yes. Yes. Okay.

Avishek Banik: Okay.

Nishant Agrawal: Makes

Avishek Banik: I used I Yeah, go ahead. So,

Ayush Singh: Mhm.

Avishek Banik: please add these things

Nishant Agrawal: sense.

Avishek Banik: up.

Ayush Singh: This thing know like uh the small paragraph will appear here when we scroll

Avishek Banik: Yeah, small paragraph here.

Ayush Singh: through move the

Avishek Banik: Yeah, small paragraph here and with the compounds with the compounds directly like link either

Ayush Singh: curve.

Avishek Banik: if uh with popc. Okay.

Ayush Singh: Okay, I can add that easily.

Avishek Banik: So if you if you are clicking over here it will go to popcam and for the gene gene

Ayush Singh: Yeah.

Avishek Banik: sequences try to uh try to put the strain name accession number or else what you can do you can do the hyperlink when you will click here it will directly redirect you to the sequence in the NCBI.

00:17:02

Ayush Singh: Okay.

Avishek Banik: Okay.

Ayush Singh: Sure.

Avishek Banik: So then then you need not to put the uh the strain number and all the things it directly

Ayush Singh: Yeah.

Avishek Banik: connected to the NCBI uh website.

Ayush Singh: Right. Works.

Avishek Banik: Okay.

Ayush Singh: Yeah.

Avishek Banik: Yeah.

Ayush Singh: So similarly for all the targets I have done the docking like all the 12 targets with based on that 24 lag we had. So we had around 288 uh so here all the data that we currently see is cached

Avishek Banik: Yeah.

Ayush Singh: like we have done the docking we saved all the files here and we had saved that loc like here inside in the front end itself like all that files that we got. If you want to see that we have that file available here. So for each uh this docking So here here the delay data data is for each dogging for each pair like for this target LR we get we dogged it with all the 24

Avishek Banik: That's that that's great.

Ayush Singh: lians.

00:18:14

Avishek Banik: That's great, man. That's great, man. Click click click any of the compound.

Ayush Singh: Uh uh yeah so for for any other compound we get around

Avishek Banik: Click any any of the compound.

Ayush Singh: list 11 files.

Avishek Banik: Okay. So,

Ayush Singh: So one is the div doc post the number one no first is the div doc post that we got the number

Avishek Banik: what are these codes? So, for two.

Ayush Singh: one post was from the div doc div doc I model that we had then the second file is the div

Avishek Banik: Okay. Okay.

Ayush Singh: doc score so all the rank from the div doc model rank the top

Avishek Banik: Great. Great. Great. Great. I understand. Great. Yeah.

Ayush Singh: 10 ranks are here similarly we got the evidence passport here so we or the next validation step

Avishek Banik: Okay.

Ayush Singh: based on the score that we generated here if the score is too low and what is the score here we got we But uh uh so it's

00:18:55

Avishek Banik: Yeah. Here you have the popcam C. See?

Ayush Singh: uh here for this uh this uh target this is lagant this C I PK C

Avishek Banik: Yeah. So you need to do you need to do hyperlink with that thing only.

Ayush Singh: I okay okay I I got that I have that idea

Avishek Banik: That's it.

Ayush Singh: about that here for then the there's this interaction parser B on the bonds and everything the

Avishek Banik: Yeah.

Ayush Singh: interaction that we got here is the receptor file and the blunt file these are the

Avishek Banik: A great man. Great. Yeah.

Ayush Singh: two different files that we got from Pupcam itself and then the interaction

Avishek Banik: Yeah. Yeah. Yeah.

Nishant Agrawal: What?

Avishek Banik: Yeah.

Ayush Singh: report it's for that B hydrogen

Avishek Banik: Just just wait let me let me check type hydrogen bonds.

Ayush Singh: bonds

Avishek Banik: Uh so receptor chains G E these are these are what? F receptor

Ayush Singh: these one are I got this from the models itself like I have no proper idea about all

00:19:43

Avishek Banik: chain

Nishant Agrawal: Which do or autod do wina? Which model output it is? This

Ayush Singh: This is from I think uh uh this one

Nishant Agrawal: one.

Avishek Banik: the the residue these are amino acid arginine its position is 112 hey

Ayush Singh: autodoc

Avishek Banik: shahu hey receptor chain E what does it amino acid who

Nishant Agrawal: I think he's

ANIRBAN SHAOO: Yes sir. Yes sir. Yes sir. Yes

Avishek Banik: hey what are these receptor chain A receptor chain Ah what does it

ANIRBAN SHAOO: sir.

Avishek Banik: mean

ANIRBAN SHAOO: Let me see. Uh

Avishek Banik: because these receptor residue these are phenile alanine and position is seven but what are

ANIRBAN SHAOO: once.

Avishek Banik: these Total age bond six and total hydrophobic it's a it's all right but what is this we need to clarify all these things up

ANIRBAN SHAOO: Um I also actually don't know about this uh receptor ENF what it is showing.

Avishek Banik: I was just Look into this. What is this? I need this

Ayush Singh: Yes, I I'll tell you about it.

00:21:02

Ayush Singh: And then we have this mechanism graph that has all the generated nodes and the

Avishek Banik: information.

Ayush Singh: edges saved here.

ANIRBAN SHAOO: It can be a bit.

Ayush Singh: This is the target here and this is the lian and then the mechanism graph we

Avishek Banik: Yeah.

Ayush Singh: have for the nodes and the edges we have all this I have no

Nishant Agrawal: He created a knowledge graph also.

Ayush Singh: idea about all this

Nishant Agrawal: He plotted a knowledge graph in the UI. The interactions

Ayush Singh: yeah then the validation priority report that we got the the priority score that we have so based on the affinity and then the confidence score and then the H bond and

Avishek Banik: Okay. Okay.

Ayush Singh: hydrogen we got the out of 100.

Avishek Banik: Yeah. Yeah. Yeah. Yeah. It's good. Good. Good. Good.

Nishant Agrawal: this score we need to

Avishek Banik: Good.

Ayush Singh: But Nishans told Nishansa told that we have to do some add some weights to this like

00:21:45

Nishant Agrawal: change

Ayush Singh: if uh like we you

Nishant Agrawal: more than the weight I will I will do a little bit of more data science in this let let's

Ayush Singh: just like as you discussed no on

Nishant Agrawal: see yeah I'll go a little bit deeper into the encoder

Ayush Singh: what basis are we giving the polarity.

Nishant Agrawal: decoder part itself But let's see that first let all the data be reviewed and validated then we will make our own score with the help of a bishek inputs and

Ayush Singh: Yeah. So here is all the evidence strength and uh this these things are generated by the AI itself.

Nishant Agrawal: important

Ayush Singh: Then we have the autodoc vinas post the here you see the affinity score is here.

Nishant Agrawal: file.

Ayush Singh: Yeah p lock file and then here is the uh vinas uh official record that we

Nishant Agrawal: Yeah.

Ayush Singh: got.

Avishek Banik: Yeah, it's all right.

Ayush Singh: So the mode one we got the so for

Avishek Banik: All

Nishant Agrawal: So think about for everything we have this for every

Ayush Singh: each so like if you see there are

00:22:34

Avishek Banik: right.

Ayush Singh: 288 uh entries now.

Nishant Agrawal: entry.

Ayush Singh: So for every target we got around 11 files. So for this I think it would be more than 2,000

Avishek Banik: Yeah.

Ayush Singh: files.

Avishek Banik: When when like uh we are we will launch it we will launch it with a paper like uh all these analysis will be will be in a paper and then like like uh have you seen the alphapold paper? They have done this similar kind of work for years and then they published the alpha pool. So we will do the similar thing. Yeah, it it looks

Nishant Agrawal: Uh yeah. So,

Avishek Banik: good.

Nishant Agrawal: so Abishi next step is to just to validate I think Anan you have validated the data right for one of the uh bacteria right? What? What?

Avishek Banik: Which weightlap validation then

Nishant Agrawal: No, not that this initial that.

ANIRBAN SHAOO: Uh actually actually I want to say something regarding this uh the work which he has done

Nishant Agrawal: Yeah.

ANIRBAN SHAOO: regarding the 12 lians and the targets there sorry yeah 24 lians and the targets were 12 uh 12.

00:23:55

Ayush Singh: H.

ANIRBAN SHAOO: So I what I could have what I have seen is that For different like target the scores are very much different means the top priority compounds are changing constantly for all the like all the targets. So it is very it is becoming very difficult for me to like understand like which which five potent lians I need to choose. So,

Nishant Agrawal: So,

Avishek Banik: Oh,

ANIRBAN SHAOO: so that so,

Nishant Agrawal: so that is the right right that is the right question you raised on

ANIRBAN SHAOO: so,

Nishant Agrawal: one.

Avishek Banik: is it is it hallucinating?

ANIRBAN SHAOO: so this is, yeah,

Avishek Banik: Is it hallucinating?

ANIRBAN SHAOO: the result is actually hallucinating and it can only be solved if there is some kind of parameters where all the scores have been displayed and if it can be like if the if there is some kind of algorithm to choose five top priority lens which can work on this entire three different bacteria means he what he has done is that for each receptor it is showing mean each like I'm talking about each uh like target is showing top pri priority compounds is becoming different like the 24 compounds we are

00:25:10

Ayush Singh: Huh?

ANIRBAN SHAOO: chosen it is becoming different so it is become very difficult for me like in one it is showing that nimolite is the highest priority in another it is showing as the directing is the first one.

Ayush Singh: You

ANIRBAN SHAOO: It's the top priority. So it's becoming very difficult for me to understand like it's difficult for him to also understand that which will be the priority. So if there is some kind of algorithm which can choose that which are the top five prior compounds which can be worked which has some

Avishek Banik: So, so anyone you want a multi-dimensional analysis

ANIRBAN SHAOO: model. Yes. Yes. That is what I am talking about like if by algorithm it if it can be

Avishek Banik: This

ANIRBAN SHAOO: solved like these are the five targets these are the five compounds which are the final like which can have moderate effect or it can have a good effect against these kind of pathogens these three different bacterias

Nishant Agrawal: So, so same question right Anurban and Amishek right what do you mean by hallucinating because you're using automina right which is a kind of a bioinformatics tool right uh div is a tool which is based on a decision technique so uh do we need to is this one of the tool data is hallucinating like the AI

00:26:28

ANIRBAN SHAOO: No, no, no,

Avishek Banik: I I no no no no no no.

ANIRBAN SHAOO: no.

Nishant Agrawal: tool.

Avishek Banik: C can you do a consensus of these two tool like uh whenever in statistics we used to take the average value. So c can can we do the consensus of this tool?

Nishant Agrawal: So how how what do you mean by consensus? I wish

Avishek Banik: Consensus means say for example in every tool they are giving some kinds of points means in server one I am ing. 95 in server two I am getting 70. So from there I can can I draw my score? It's a $90 + 70$ divided by two like that.

Nishant Agrawal: But is this a right approach?

Avishek Banik: I don't know. I'm just asking.

Nishant Agrawal: I think it will be a kind of a

ANIRBAN SHAOO: sir I can say something about this okay okay are you uh please listen

Avishek Banik: Yeah. Yeah. Go ahead. Go ahead. Go ahead.

Ayush Singh: Yeah.

ANIRBAN SHAOO: to this let's say uh let's say one compound like like I'm giving an example like

00:27:24

Ayush Singh: Yeah.

ANIRBAN SHAOO: nimolide okay so nimolide in one one target is showing that

Ayush Singh: Mhm.

ANIRBAN SHAOO: it's a top priority okay it's a In some other target it is showing like let's say it is

Ayush Singh: Yeah. Yeah.

ANIRBAN SHAOO: a 12th priority compound like this it is showing different priorities do a mean value of it what I'm talking

Ayush Singh: H.

ANIRBAN SHAOO: about the like the confidence score if you can do a mean value of this nimolide total against this then we can we will get a average score of everything every compound we can get like this

Ayush Singh: No, no.

ANIRBAN SHAOO: Right?

Ayush Singh: Like main value of what? Membolite. Like how do we do that?

ANIRBAN SHAOO: Means means I'm telling you for example let's uh say the nimolite is having a value of let's say a pqrs something that was uh receptor there it is showing that it has a confidence level of around

Ayush Singh: Mhm.

ANIRBAN SHAOO: 90%.

Ayush Singh: Okay.

ANIRBAN SHAOO: Okay in another in another target it is showing that it is having a value of let's say

00:28:26

Ayush Singh: Okay.

ANIRBAN SHAOO: 50%. Right?

Ayush Singh: Okay.

ANIRBAN SHAOO: So,

Ayush Singh: Yeah.

ANIRBAN SHAOO: so like this you will get like the 12 the 12 targets you will get a different type of scores you have got a different type of scores right for each compound then what you do then you have to divide that by 12

Ayush Singh: Yeah.

ANIRBAN SHAOO: or something like that like because there are 12 targets if you take the average score you will get divide by 12 and then if you can present the value for this means like you will get a value like for nimolite one okay such then you will get another

value for like another compound compound two, compound three, compound four. So this might actually solve kind certain kind of the problem to target mean say the

Avishek Banik: Yeah, it's a kind of same thing like it's a kind of taking average.

ANIRBAN SHAOO: highest

Avishek Banik: It's a kind of median value that we are talking about.

Nishant Agrawal: Okay. Across the two tools, right? Across the autodoc and uh div doc, right?

00:29:30

ANIRBAN SHAOO: Yes. Yes. Yes sir.

Avishek Banik: Yes.

ANIRBAN SHAOO: Yes sir.

Avishek Banik: Yes. Yes.

ANIRBAN SHAOO: Actually I was talking about that only.

Nishant Agrawal: But

ANIRBAN SHAOO: So if

Ayush Singh: But are you are you are you like like one thing I want to ask are you thinking this because of the priority

ANIRBAN SHAOO: if

Ayush Singh: score that we got getting or these score priorities for we still

ANIRBAN SHAOO: no see let's let's say let's say I'm telling you like you got nimolide in one uh

Ayush Singh: think uh

ANIRBAN SHAOO: receptor you are getting that in one target you are getting in the uh in the number one c at the one position but in another

Ayush Singh: uh that's why that got that that I understood but that we got number one that we are getting is based on upon

ANIRBAN SHAOO: uh priority score.

Ayush Singh: the priority score that we got the yeah the affinity score and these

Nishant Agrawal: live.

ANIRBAN SHAOO: Yes. Yes. Yes.

Nishant Agrawal: Right.

Ayush Singh: things I think they are similar to the other

00:30:08

ANIRBAN SHAOO: You have to ha but you have to also do it for the other you have to

Ayush Singh: targets.

ANIRBAN SHAOO: also take the consideration of this nimbo light for other uh like targets also. So you have to get an average right because until and unless we get

Ayush Singh: Mhm.

ANIRBAN SHAOO: the proper like we cannot screen the entire 24 compounds while while we are doing wet lab we have to take

Avishek Banik: uh on only one look whatever I am telling and you are telling almost the same thing but uh what I am asking is it wise to take the average or we need to think and

ANIRBAN SHAOO: h

Avishek Banik: then we need to decide.

Nishant Agrawal: But my question is both models are giving different uh uh units

Avishek Banik: Yes. Yes.

Nishant Agrawal: outputs.

Avishek Banik: Because the priority and the backup of these models are different

Nishant Agrawal: Yes. So how do we how do we do that like average and all because we need to normalize it right both of

Avishek Banik: but yes yes yes yes.

ANIRBAN SHAOO: Yes.

00:31:07

ANIRBAN SHAOO: Yes. So that we have to

Nishant Agrawal: the

Avishek Banik: So normal norm normalization can be done with the most probably any

ANIRBAN SHAOO: do

Avishek Banik: wet lab experiment.

Nishant Agrawal: okay

ANIRBAN SHAOO: sir. Uh uh then we have to take more than five

Nishant Agrawal: but huh but but the question is Aisha it's a chicken and egg problem right we need to define the wet lab experiments

ANIRBAN SHAOO: compounds.

Nishant Agrawal: based on this to try it So we can narrow down

Avishek Banik: No. So, so, so it look what I am thinking I I have already thought about this. So,

Nishant Agrawal: and

Avishek Banik: there will be versions update will come. So, currently go as it is. go as it is with some average or something like that. Let us give some time to think whether we will go for average or any kind of algorithm is there. So that will be the version one. So when we have the data then we can come back to it do the normalization and then we have the version two.

00:32:03

Avishek Banik: It will be like that update automatic update. We will do updating and the normalization like uh for uh look there are two kinds of normalization can be done. When we used to do metabolomic experiment one is called internal standard and

another is called standard inside the machine. Okay. So for if I know what I am targeting I use internal standard in the sample. If I don't know what I am targeting then my normalization happens based on the internal standard given by the machine. So how machine is calibrating based on that the system is getting normalized. Is it clear to you?

Nishant Agrawal: I'm thinking, can you say it again?

Avishek Banik: Two things are there. Look, look at me. One thing is known, another thing is unknown. In known what we used to do, say for example,

Nishant Agrawal: Okay.

Avishek Banik: five compounds are there and I am adding another compound like this in the mixture. What is not present in that mixture? Okay. Then what I am and then so the these are all unknown.

00:33:25

Avishek Banik: These the five mixtures these are unknown. But this is known to me. I know its molecular weight and how much I am adding into this system. So from the molecular weight I can calculate.

Nishant Agrawal: Okay.

Avishek Banik: It's basically the abog number and how much I am giving. Okay. And now I'm putting in into the system and running the system. So then then what I will do I will normalize these five compounds with

Nishant Agrawal: Okay.

Avishek Banik: this one. I will simply deduct this one and things will be normalized based on the this. So this is called internal standard. So this will be done when we have the weightlap data. Now currently we do not have the weightlap data.

Nishant Agrawal: Okay.

Avishek Banik: So we have this thing and here inside machine used to calibrate itself. When we machine used to calibrate itself. So here we have to rely on the machine calibrate

Nishant Agrawal: Okay.

Avishek Banik: calibration because we don't know what to use over here what is the suitable thing over here you understand what I am trying to say so here we will write in the paper in the discussion that it is normalized like that in that future prospect we will do wetlap experiment and then it will be normalized based on the wet lab experiment so it will be our version two like So give us some time and

00:34:50

Nishant Agrawal: Okay. Okay.

Avishek Banik: we need to think about this thing because this is very critical thing.

Nishant Agrawal: Yeah.

Avishek Banik: How to formalize that thing whether we will take the average value or something we need to

Nishant Agrawal: Something. Yes. Yes.

Avishek Banik: consider.

Nishant Agrawal: And u like um I I usually share the outcome units for both uh in the chat group so that Aneran and I already know Abhishek can also think about it right and we can decide this.

Avishek Banik: Yeah.

Nishant Agrawal: So so minutes of the meeting right?

Ayush Singh: Okay.

Avishek Banik: Minutes of the meeting.

Nishant Agrawal: Yeah. So, so basically Abishek if you see this demo um how do you think like uh anything like on

Avishek Banik: Yeah.

Ayush Singh: Yeah.

Nishant Agrawal: the AI side or bioinformatics or the you before the vet lab side

Avishek Banik: No, this looks good.

Nishant Agrawal: uh any really

Avishek Banik: This looks really good. I has worked very hard.

Nishant Agrawal: good?

Avishek Banik: Aayush and Arian, it's a commendable job. like whatever the input that we are giving you have made it.

00:35:44

Nishant Agrawal: Yeah. Yes.

Avishek Banik: So we are very close but uh until or unless we are doing weight lab experiment uh

Ayush Singh: Thank you.

Avishek Banik: we cannot say anything but but at least we will have some hits from there and I wish like an yesterday I I told uh the same

Ayush Singh: Yes,

Nishant Agrawal: Yes.

Ayush Singh: sir.

Avishek Banik: thing I'll I'll ask you like we need to do the comparative thing like uh server one is giving that server uh two is giving that now if you are combining server one and server two so what is your advancement over the pre-existing knowledge that you are getting from this uh whatever you have prepared You understood what I'm trying to say.

Nishant Agrawal: Basically unique unique uh what is your

Avishek Banik: What is the uniqueness of your that UI that you have

Nishant Agrawal: mode?

Avishek Banik: prepared?

Ayush Singh: Okay.

Avishek Banik: How how you are taking it forward?

Ayush Singh: This is

Avishek Banik: How you are taking it forward?

Ayush Singh: like this UI.

Avishek Banik: how vabrian bio AI is different from the other pre-existing uh that uh binformatic tools

00:36:59

ANIRBAN SHAOO: Actually that will be also a great thing for me because I will also include that in the slide that why to go for the mereon AI computational tool

Ayush Singh: Mhm.

ANIRBAN SHAOO: analysis.

Avishek Banik: that that that's why that's why I'm telling that's why I'm telling at the end of the day you need to find a research

ANIRBAN SHAOO: Uh

Avishek Banik: gap and Now the next step is you need to fulfill the research gap. So how you are fulfilling the research gap? I wish I need your help regarding this. You understood?

Ayush Singh: So some

Avishek Banik: So so this is one server. This is another server. Okay. So this has a specific algorithm.

Ayush Singh: Mhm.

Avishek Banik: This one has a specific algorithm. So what you are doing? You are combining these two and giving a new direction.

Ayush Singh: Yeah.

Avishek Banik: So what is its gap? What is its gap?

Ayush Singh: Yeah.

Avishek Banik: And now how you are fulfilling these two gap to a new

Nishant Agrawal: Yeah, server could be a model here for you.

00:37:53

Avishek Banik: direction.

Ayush Singh: Yeah. That's what I understood.

Nishant Agrawal: uh you just basically that what is the correlation that we are doing which

Ayush Singh: That's what I understood. the whole thing.

Nishant Agrawal: is not obvious right if I put it that

Ayush Singh: Yeah. Yeah. Yeah.

Avishek Banik: Yeah.

Ayush Singh: For I think for for uh for us the hero thing was the priority score because based on

Nishant Agrawal: way

Ayush Singh: that we were determining uh is it we should consider it for wet lab validation or we should continue with this thing or no because we got two different scores. If you see here we get the affinity score is this is the affinity score and this is the dip confidence score. So these two models were good in their places. They gave that uh scores to the to us. But what we have done is we have added all the 12 targets and lens. We can if you see here's the test a custom compound as well.

Avishek Banik: That is that yeah that is good.

00:38:44

Avishek Banik: That is good. So none have walk in that direction.

Ayush Singh: Mhm.

Avishek Banik: So we are looking at a bigger picture.

Ayush Singh: Exactly.

Avishek Banik: It's a holistic view. It's a bird's eye view.

Ayush Singh: Yes.

Avishek Banik: So we are looking the problem from the top.

Ayush Singh: Yeah. Stop.

Avishek Banik: And if you are looking the problem from the top the periphery is much more.

Ayush Singh: Yeah.

Avishek Banik: So that is good. No one is looking in this direction. So that that will give us a great advantage.

Ayush Singh: Yeah.

Avishek Banik: Okay.

Nishant Agrawal: So I have answer for this Aishek uh just to add here right what is the no novelty right

Avishek Banik: Yeah.

Ayush Singh: Mhm.

Nishant Agrawal: like what we novelty could not be building the bionformatics tool or the AI

Avishek Banik: Yeah.

Nishant Agrawal: model but the correlation right let's say we are bringing the piece together to get the analysis for to decide on the wetlab part okay the second part is from Each one the JSON the SDF and other file he has done for 2000 file we can draw some very good inferences out of

00:39:34

Avishek Banik: Yeah.

Nishant Agrawal: it for the new for the new compounds and for the new learning that

Avishek Banik: Yeah.

Nishant Agrawal: okay what is the different combination that have not been tried

Avishek Banik: Write this thing.

Nishant Agrawal: yet

Avishek Banik: I give give your uh give your email id in the box.

Ayush Singh: Yes, sir.

Avishek Banik: So what I will do I will share a word document start writing all these things the material

Ayush Singh: Okay.

Avishek Banik: method the whatever this Nishant has whatever he has added just write these things

Ayush Singh: Uhhuh.

Avishek Banik: up okay so that we'll be in all will be in the same page and

Ayush Singh: Okay. I

Avishek Banik: okay so uh this is your email id right okay put it in the group put it in the

Ayush Singh: Yeah.

Avishek Banik: WhatsApp group so that in night I I'll come again and I'll like start working

Ayush Singh: Okay.

Avishek Banik: for my my work I will start at like at 11:00 so then I will share all those

ANIRBAN SHAOO: What's

Avishek Banik: documents before that I'm not

00:40:29

ANIRBAN SHAOO: that?

Avishek Banik: getting time for myself

Nishant Agrawal: So,

Avishek Banik: yeah

Nishant Agrawal: so basically um are you pull the all the Gemini notes and create a very

ANIRBAN SHAOO: Excuse

Nishant Agrawal: short top five or top 10 notes also which you can share.

Ayush Singh: I'll add that summary as

Nishant Agrawal: Yeah.

Ayush Singh: well.

Nishant Agrawal: So work on that summary immediately after the meeting so you'll not get a you lose a con context whatever we are

ANIRBAN SHAOO: me.

Ayush Singh: Yeah.

Nishant Agrawal: talking because sometimes don't understand.

Avishek Banik: It's It's a great thing.

Nishant Agrawal: Okay.

Avishek Banik: It's a great thing. We need to

Nishant Agrawal: Yes.

Avishek Banik: validate.

Nishant Agrawal: Yes. So,

Ayush Singh: Yeah.

Nishant Agrawal: so uh Abishek then I'll talk to you then and you talk to Aniran also how we can whatever we have today how we can sell it.

Avishek Banik: Yeah.

Nishant Agrawal: You understand what I'm saying in the idea idea for the daily meeting.

Avishek Banik: Yeah. Yeah. Yeah. Yeah, this looks good.

Nishant Agrawal: Yeah.

00:41:22

Avishek Banik: It will be sellable.

Nishant Agrawal: So yeah that pitch that pitch is still I'm I'm

Avishek Banik: I'm telling you, it will be sellable. But

Nishant Agrawal: just I need to calibrate with you. Maybe I'm not getting that uh how to deliver that because that's where I'm just trying to

Avishek Banik: yeah.

Nishant Agrawal: correlate the the the I'm thinking that I'm the the principal secretary and the scientist who will be listening this that how I will be convinced for this. So that and I think we are closer but the closing statement we need to

Avishek Banik: Yeah, definitely.

Nishant Agrawal: add.

Avishek Banik: Definitely the the way these guys has put all things up. Uh this this this looks very promising to me.

Nishant Agrawal: Yes.

Avishek Banik: like from here we can build uh something.

Nishant Agrawal: Yes, this can go to the very different level of tools.

Avishek Banik: Yeah. Yeah.

Nishant Agrawal: We can extend this to a very level not just for Aush but it

Avishek Banik: Definitely definitely definitely definitely

Nishant Agrawal: can go for precision.

Avishek Banik: anything anything you can do anything you can

00:42:13

Nishant Agrawal: I think this can also link to the fizadorium part also because this technique

Avishek Banik: do fuserium fuserium part and in

Nishant Agrawal: in

Avishek Banik: in future if someone look it's a protein liant interaction forget about this bacteria and all

Nishant Agrawal: Yeah.

Avishek Banik: these things it's a protein liant interaction you can sell this thing to the

Nishant Agrawal: Yeah.

Avishek Banik: the cancer guys also like they are looking for different kinds of drugs

Ayush Singh: Currently like currently we are done with just for the aush compound we can expand it to other compound as

Avishek Banik: and it's anything

Nishant Agrawal: Yes.

Ayush Singh: well.

Avishek Banik: anything it's a base. So here think about think like this it's a protein and liant interaction.

Ayush Singh: Yeah.

Avishek Banik: So it can be anything it can be anything anything first need to

Ayush Singh: Right.

Nishant Agrawal: Anything yet?

Avishek Banik: optimize with the bacterial system then then like if it's working in

Ayush Singh: Yeah.

Nishant Agrawal: Yeah,

Avishek Banik: the weightlap condition I'll be mad.

Nishant Agrawal: it will big big help for farmers by the

00:43:05

Avishek Banik: No it's a yeah it's it's a big thing. It's a big thing.

Nishant Agrawal: way.

Avishek Banik: People used to do like 10 years of research with this and now look with this this much of compounds that I did it. It takes how how many days we are discussing it's more than four months I guess. So it take four months bionformatically to put up all these things.

Nishant Agrawal: Okay.

Ayush Singh: Look.

Avishek Banik: Think about one guy is working in a wetlab situation. It will take minimum five to six years. what Aayush and Aniran has done in 4 months. It's it's not a joke because wet lab it's a completely all different ball game

Nishant Agrawal: Yes.

Avishek Banik: together.

Nishant Agrawal: And u another thing Abishek in my mind before I forget is do you want to have ESM fold for the prediction for the wet lab we know the ESM fold I think we ran right

Avishek Banik: Uh

Nishant Agrawal: Aush we ran for the other chemotherapy and other thing right we have the model ready do you want

00:44:07

Ayush Singh: Yeah.

Nishant Agrawal: to try but what could be the input and what could be the output

Avishek Banik: yeah

Ayush Singh: No, but but sir ESM code is sim just like that alpha fold. No, like the collab fold that we had like we just had add that uh sequence to them and

Nishant Agrawal: Yes.

Ayush Singh: it generates the uh the

Nishant Agrawal: The fold.

Avishek Banik: structure.

Nishant Agrawal: See the

Ayush Singh: structure

Avishek Banik: So yeah. So,

Nishant Agrawal: Yeah.

Avishek Banik: so alpha fold is much more convincing in our system.

Ayush Singh: better.

Avishek Banik: Let it be.

Ayush Singh: Yeah.

Avishek Banik: Let it be. Okay. So, uh like it will be a replacement of this currently at this stage. It's not necessary.

Nishant Agrawal: not necessary.

Avishek Banik: Yeah. Not

Nishant Agrawal: So after this course you don't want to do any more processing for which can give more edge

Avishek Banik: necessary.

Nishant Agrawal: value. It's it is enough whatever h bonds and scores confidence scores and we are getting that is enough

00:44:58

Avishek Banik: Uh yes uh Anan take a note of this.

Nishant Agrawal: Okay.

Avishek Banik: We need to discuss in two ways like two

ANIRBAN SHAOO: Sir uh sir uh one thing is that also uh we are talking about the

Avishek Banik: things.

ANIRBAN SHAOO: uh hydrogen bonding also we need to see the electrostatic interactions also if the compound if the compound has some ability like some positive effect uh you understand sir so that like a like if there's some amin group or something like

Avishek Banik: Uh,

ANIRBAN SHAOO: that so if that is somehow helping the like it can be much more helpful because it will like work more effectively. We have to also consider not only the hydrogen bonding but also we can consider the electrostatic interaction other bonds the coalent bonds I feel. So

Nishant Agrawal: So we can so so Aneran what you're suggesting is to run some more computation and to validate the results after this.

ANIRBAN SHAOO: yeah. Uh the thing is that he he has done excellent about this hydrogen bondings and all the weak bonding like which happens. But I also want to see the strong interactions like the covalent interaction whether there is some kind of covalent interaction which can be formed or whether there is some electrostatic interactions which can happen

00:46:19

Nishant Agrawal: So okay.

ANIRBAN SHAOO: explain I'll explain to him what I am talking about but I need to also

Nishant Agrawal: Okay.

ANIRBAN SHAOO: ask sir sir whether this approach is good or not like sex I'm

Avishek Banik: You can add look look you can add but here mostly non-covalent interaction

ANIRBAN SHAOO: talking

Nishant Agrawal: Okay.

Avishek Banik: is much more uh like prominent than the covalent

ANIRBAN SHAOO: Sir,

Avishek Banik: interaction and Yeah,

ANIRBAN SHAOO: I am I actually I was talking about this positive negative thing. You

Avishek Banik: I I can understand. I can understand. But the thing is that look adding much more things are

ANIRBAN SHAOO: understand?

Avishek Banik: always good but you have to justify so why we doing this and

Nishant Agrawal: Just Yeah. Why we doing

Avishek Banik: another thing can we have any kind of I compound which has any three group I don't think

Nishant Agrawal: this?

Avishek Banik: so these are all phytochemicals these are all phytochemicals

ANIRBAN SHAOO: I need to Yes,

Avishek Banik: right so the does any

ANIRBAN SHAOO: sir. Yes, sir.

00:47:13

Avishek Banik: phytochemical have like those kind coalent interaction I don't think

ANIRBAN SHAOO: I have to just check on that.

Avishek Banik: so check it check it look look you can add

ANIRBAN SHAOO: I'll I'll check today also.

Avishek Banik: it anytime but if you are adding another parameter so it can drag to our experiment in any other direction so if it's a necessary so what we can do we can use a checkbox coalent interaction non-coalent interaction

ANIRBAN SHAOO: H.

Avishek Banik: and then the user can uh define uh Nishant like yesterday we have

ANIRBAN SHAOO: Yes.

Avishek Banik: discussed like uh we can do like two check box. So whatever we are considering these are all non-coalent interaction and if

Nishant Agrawal: No. Okay.

Avishek Banik: If some user want to do any kind of coalent interaction, he needs to do the checkboxing and then it's a customized thing that we can add.

Nishant Agrawal: Okay. So

Avishek Banik: So one will be in default mode and another will be in the customize mode.

Nishant Agrawal: default.

Avishek Banik: In default mode checkbox will be different. In customize mode checkbox will be

00:48:18

Nishant Agrawal: Okay. Right. So,

Avishek Banik: different.

Nishant Agrawal: so what type of models and other things can be run or what type of processing so that Aush can implement this quickly because it not take much time to implement it.

Avishek Banik: Give us give us a couple of days. I'll let you know these two

Nishant Agrawal: Okay. Okay. So that's once you give a green signal both of you we can implement that part as a checkbox

Avishek Banik: things.

Nishant Agrawal: option and get that additional data what actually ana are saying to just correlate and more more future proof.

Avishek Banik: Okay. Okay. Okay.

Nishant Agrawal: Okay.

Avishek Banik: Okay. So, so you will you will present both of these to Swiss Swiss

Nishant Agrawal: So

Avishek Banik: guys. Okay.

Nishant Agrawal: both and I'm planning to fly

Avishek Banik: Okay. What?

Nishant Agrawal: uh mid of August or something. I am waiting for some this aush meeting. I'm delaying my tickets all the time but uh but

Avishek Banik: Go.

00:49:10

Avishek Banik: Go. Go for one. Go for one. At least we will have

Nishant Agrawal: yeah so I will come to Kolkata before I fly.

Avishek Banik: something.

Nishant Agrawal: I'll come to Kolkata before I fly.

ANIRBAN SHAOO: sir

Avishek Banik: Okay.

Nishant Agrawal: So sing for the deck and all.

Avishek Banik: Okay. Okay. Okay. Okay.

Nishant Agrawal: Okay.

ANIRBAN SHAOO: sir check the WhatsApp once uh I have sent you in the IO group

Nishant Agrawal: You send something. Just a second. What is this?

Avishek Banik: coalent. Yes, phytochemicals utilize both coalent and electrostatic interaction. Then add it. Then add it. Look interaction are there but these are not prominent enough. These are not prominent enough. You can add it.

Nishant Agrawal: Oh.

Avishek Banik: Look Anan always you can add it.

Ayush Singh: Watch. Watch out.

Avishek Banik: No no no no issues are there but it should not take our uh like interpretation to another direction. So if it's not necessary it's better to check it out.

Nishant Agrawal: Yeah. So,

ANIRBAN SHAOO: Okay.

Nishant Agrawal: everyone Sleep over it.

ANIRBAN SHAOO: Okay.

Avishek Banik: Okay.

ANIRBAN SHAOO: Okay.

Nishant Agrawal: Sleep over this thought and talk to Vish.

Avishek Banik: We we will discuss and let you know.

Nishant Agrawal: Yeah.

ANIRBAN SHAOO: Okay.

Nishant Agrawal: Because if it is required,

Avishek Banik: Okay.

ANIRBAN SHAOO: Fine.

Nishant Agrawal: we will do in a day.

ANIRBAN SHAOO: Yes.

Avishek Banik: Sure. Sure.

ANIRBAN SHAOO: Yes.

Avishek Banik: Sure.

Nishant Agrawal: Okay.

Avishek Banik: Okay.

ANIRBAN SHAOO: Yeah.

Avishek Banik: Okay, man. Yeah.

Nishant Agrawal: All right, guys.

Avishek Banik: Okay. Bye.

Nishant Agrawal: I think I'll call you later.

Avishek Banik: Bye. Yeah.

Nishant Agrawal: Yeah.

Ayush Singh: Bye sir.

Nishant Agrawal: Bye, guys.

Ayush Singh: Thank you.

Transcription ended after 00:50:41

This editable transcript was computer generated and might contain errors. People can also change the text after it was created.